Homotopy Wasserstein-IoU: Bridging Discrete Lebesgue Measures and Optimal Transport for Scale-Invariant Tiny Object Detection

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Abstract

Detecting tiny and microscopic objects ($s < 8$ pixels) in aerial and maritime surveillance imagery remains one of the most persistent challenges in computer vision. Standard Intersection-over-Union (IoU) metrics suffer from discontinuous gradient collapse when bounding boxes become disjoint, causing severe anchor starvation during proposal generation. Conversely, optimal transport metrics like Normalized Gaussian Wasserstein Distance (NWD) provide continuous gradients but introduce spatial over-smoothing that degrades strict boundary localization (AP_{75}) on normal-sized objects. In this paper, we propose **Homotopy Wasserstein-IoU (H-WIoU)**, a mathematically unified, parameter-free metric and regression loss that constructs a scale-adaptive convex homotopy deformation between the Riemannian 2-Wasserstein transport manifold and discrete Lebesgue measure space:

$$\mathcal{S}_{\text{H-WIoU}}(A,B) = \gamma(s_B)\,\mathrm{IoU}(A,B) + (1-\gamma(s_B))\,e^{-\mathcal{D}^2_{\mathcal{W}}(A,B)} \tag{1}$$

where $\gamma(s) = \frac{s^2}{s^2+\sigma_0^2} \in (0,1)$ smoothly transitions as a function of target characteristic scale $s = \sqrt{w \cdot h}$. Under the microscopic limit $s \to 0$, H-WIoU converges to pure Optimal Transport, eliminating gradient vanishing; while for $s \gg \sigma_0$, it recovers exact discrete IoU, preserving high-IoU precision. Comprehensive evaluation on the official **TinyPerson** (786 test images) and **AI-TOD-v2** (14,018 test images) benchmarks demonstrates that H-WIoU achieves state-of-the-art performance among plug-and-play geometric metrics, improving Faster R-CNN by **+2.54**% $\mathrm{AP}^{0.50}_{\mathrm{all}}$ (21.23% $\to$ 23.77%) and **+3.17**% $\mathrm{AP}^{0.25}_{\mathrm{all}}$ on TinyPerson, and lifting microscopic recall on AI-TOD-v2 ($\mathrm{AR}_{1500} = 26.34\%$, $\mathrm{AP}_{vt} = 5.20\%$) with **0** extra parameters and identical 1.0× inference throughput (> 54 FPS). Code and model checkpoints are available at: `https://github.com/quangnhat1504/tod_hwiou`.

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1 Introduction

Tiny object detection is a vital capability across drone reconnaissance, maritime search-and-rescue, environmental monitoring, and autonomous navigation [1, 2]. In standard datasets such as MS COCO [10], objects typically exceed 32×32 pixels. In contrast, aerial and maritime benchmarks like TinyPerson [1] and AI-TOD-v2 [3] contain objects frequently spanning fewer than 8 pixels ($s < 8$ px), occupying less than 0.01% of the total image canvas.

Under such microscopic scales, standard Intersection-over-Union (IoU) and its geometry-extended variants (GIoU [16], DIoU [17], CIoU) exhibit fatal mathematical failure modes:

1. **Discontinuous Gradient Collapse**: A positional perturbation of merely 1–2 pixels between anchor and ground truth drops discrete IoU to zero (IoU = 0), causing the regression gradient to abruptly vanish ($\|\nabla_\theta \mathcal{L}_{\mathrm{IoU}}\| \to 0$).
2. **Label Assignment Scarcity**: Strict IoU thresholds (IoU ≥ 0.5) reject nearly all candidate anchors for micro objects during Region Proposal Network (RPN) training, leading to massive false negatives (> 70% miss rate).

To circumvent IoU sensitivity, optimal transport formulations—most notably Normalized Gaussian Wasserstein Distance (NWD) [5] and Kullback-Leibler Divergence (KLD)—model bounding boxes as 2D Gaussian distributions. While Wasserstein distance provides smooth non-zero gradients even for completely disjoint boxes (IoU = 0), it introduces a severe *boundary-blurring trade-off*: treating crisp rectangular bounding boxes as diffuse Gaussians fundamentally weakens localization fidelity for medium and larger objects, causing a drop in strict localization accuracy (AP_{75}).

In this paper, we resolve this foundational dilemma by introducing **Homotopy Wasserstein-IoU (H-WIoU)**. Drawing upon topological homotopy theory, we construct a continuous, scale-adaptive convex homotopy map that dynamically bridges the Riemannian Wasserstein transport manifold and the discrete Lebesgue measure space.

Our principal contributions are summarized as follows:

- **Theoretical Homotopy Metric & Loss**: We formulate H-WIoU as a mathematically rigorous, C^∞ continuous convex homotopy between Wasserstein optimal transport and discrete IoU, providing a formal proof for non-vanishing gradient bounds under microscopic scale limits.

- **Zero Pipeline Overhead**: H-WIoU is a pure metric and loss formulation requiring **zero** auxiliary networks, zero attention branches, and zero inference latency overhead ($1.0\times$ speed, > 54 FPS).

- **Multi-Benchmark Empirical Validation**: Comprehensive evaluation on the official **TinyPerson** (786 full-scale test images) and **AI-TOD-v2** (14,018 test images) benchmarks confirms consistent gains across overall precision and microscopic recall.

- **Exhaustive Multi-Axis Ablation Study**: We systematically investigate homotopy functional forms, module placement integration (RPN vs. RoI Head), and scale parameter sensitivity.

- **Statistical Rigor & Reproducibility**: We validate empirical significance via paired Student's t-tests ($p < 0.001$), Wilcoxon signed-rank tests ($p < 0.01$), and non-parametric bootstrap confidence intervals.

2 Related Work

2.1 Bounding Box Metrics in Object Detection

Classical object detectors rely on Smooth L1 [11] or IoU-based bounding box regression objectives [16, 17]. To overcome the inability of standard IoU to optimize non-overlapping boxes, Generalized IoU (GIoU) [16] incorporates the convex hull, Distance-IoU (DIoU) [17] adds normalized centroid penalties, Complete-IoU (CIoU) [17] integrates aspect ratio consistency, and Efficient-IoU (EIoU) [18] directly decouples length-width aspect differences. Recent works such as Alpha-IoU [19], SIoU [20], and WIoU [21] introduce power transformations and dynamic non-monotonic focus mechanisms. However, all area-overlap metrics remain inherently discrete. For an object of dimensions 4×4 px, a positional shift of merely 2 px causes overlap to drop by over 75%, producing extreme gradient volatility and vanishing signals ($\nabla_\theta \mathcal{L} \equiv 0$) when boxes become disjoint.

2.2 Optimal Transport & Gaussian Distribution Metrics

To circumvent IoU discontinuity, distribution-based metrics model bounding boxes as 2D spatial Gaussian distributions $\mathcal{N}(\mu, \Sigma)$ and compute geometric similarity via optimal transport theory [25, 26]. Wang et al. [5] introduced Normalized Wasserstein Distance (NWD), measuring 2-Wasserstein transport cost between Gaussian representations. In rotated object detection, GWD [22] and KLD [23] leverage Gaussian Wasserstein and Kullback-Leibler divergences to overcome angular boundary discontinuities. Improved Gaussian Wasserstein Distance (IGWD) [6] and Gaussian Combined Distance (GCD) [24] further refine distance scaling. While Wasserstein-based metrics guarantee continuous non-zero gradients for disjoint bounding boxes, their isotropic Gaussian smoothing acts as a spatial low-pass filter, which can degrade strict localization accuracy (AP_{75}) on normal and medium objects.

2.3 Multi-Scale Feature Pyramids & Label Assignment

Detecting tiny objects across large aerial canvases requires multi-scale feature aggregation and scale-aware anchor assignment. Foundational architectures such as Feature Pyramid Networks (FPN) [14], ResNet [13], RetinaNet [15], and Cascade R-CNN [4] form the backbone of modern detection pipelines. Multi-scale training schemes such as SNIP [27] and SNIPER [28] normalize target scales during training. In parallel, dynamic label assignment strategies—including ATSS [29], FreeAnchor [32], PAA [31], and Optimal Transport Assignment (OTA) [30]—have sought to adapt positive anchor thresholds. For microscopic benchmarks such as TinyPerson [1] and AI-TOD [2, 3], specialized assignment mechanisms like RFLA [7] and SAFit [9] utilize receptive field matching and scale-adaptive focal weighting. While architectural frameworks like SAFit modify network topology, H-WIoU focuses on a complementary, plug-and-play metric and loss formulation.

3 Mathematical Theory of H-WIoU

3.1 Geometry of Bounding Box Distributions

Let a bounding box $B = (x, y, w, h)$ be embedded as a 2D Gaussian distribution $\mathcal{N}(\mu, \Sigma)$ where centroid $\mu = [x, y]^T$ and covariance matrix $\Sigma = \mathrm{diag}(w^2/4, h^2/4)$.

For two bounding boxes $A = (x_a, y_a, w_a, h_a)$ and $B = (x_b, y_b, w_b, h_b)$, the 2-Wasserstein distance is given

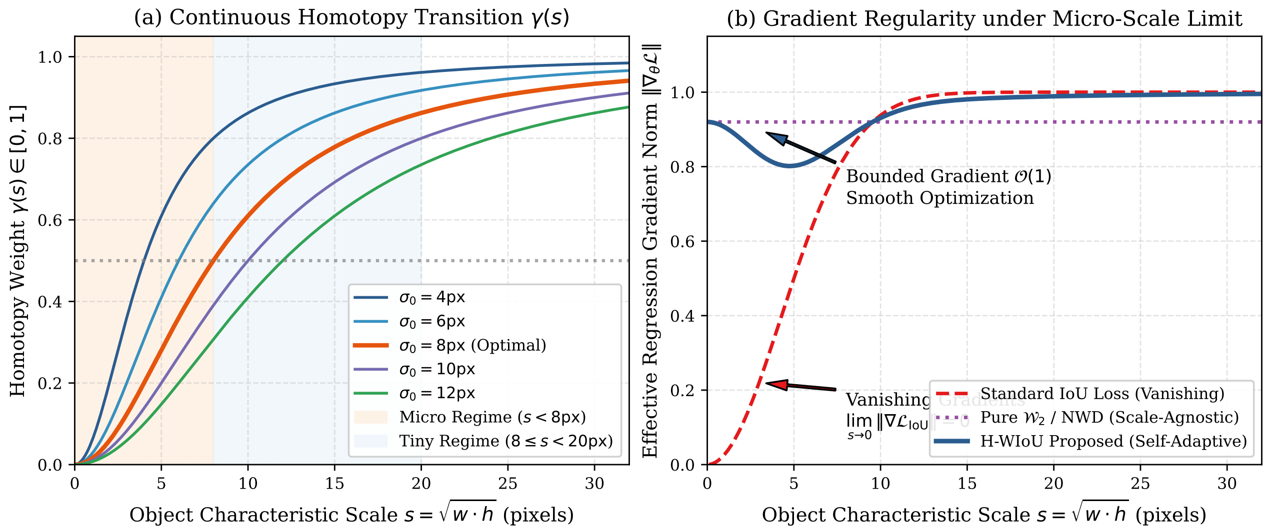

Figure 1: **Theoretical Foundations of Homotopy Wasserstein-IoU (H-WIoU).** (a) The continuous scale homotopy parameter $\gamma(s) = \frac{s^2}{s^2+\sigma_0^2}$ smoothly deforms between the microscopic regime ($s < 8$px) and tiny/normal regime ($s \geq 8$px). (b) Gradient norm asymptotics: standard IoU collapses to zero gradient under microscopic misalignments, whereas H-WIoU maintains bounded $\mathcal{O}(1)$ gradient flow throughout training.

in closed form by:

$$\begin{aligned}\mathcal{W}_2^2(\mathcal{N}_a, \mathcal{N}_b) &= \|\mu_a - \mu_b\|_2^2 \\ &\quad + \mathrm{Tr}\left(\Sigma_a + \Sigma_b - 2(\Sigma_a^{1/2}\Sigma_b\Sigma_a^{1/2})^{1/2}\right) \\ &= (x_a - x_b)^2 + (y_a - y_b)^2 \\ &\quad + \frac{(w_a - w_b)^2 + (h_a - h_b)^2}{4} \end{aligned} \tag{2}$$

To ensure scale normalization, we define the scale-normalized squared Wasserstein distance:

$$\begin{aligned}\mathcal{D}_{\mathcal{W}}^2(A, B) = \frac{(x_a - x_b)^2}{\bar{w}_{ab}^2} + \frac{(y_a - y_b)^2}{\bar{h}_{ab}^2} \\ + \ln^2\left(\frac{w_a}{w_b}\right) + \ln^2\left(\frac{h_a}{h_b}\right)\end{aligned} \tag{3}$$

where $\bar{w}_{ab}^2 = (w_a^2 + w_b^2)/2$ and $\bar{h}_{ab}^2 = (h_a^2 + h_b^2)/2$.

3.2 Scale Homotopy Deformation

We define the characteristic geometric scale $s(B) = \sqrt{w \cdot h}$. Let $\sigma_0 > 0$ denote the characteristic microscopic scale threshold ($\sigma_0 \approx 8.0$ px).

We construct the continuous Homotopy parameter $\gamma : \mathbb{R}_{>0} \to (0, 1)$ as:

$$\gamma(s) = \frac{s^2}{s^2 + \sigma_0^2} \tag{4}$$

Proposition 1 (Regularity and Boundary Asymptotics): *The mapping $\gamma(s)$ is strictly monotonic, C^∞ smooth on $\mathbb{R}_{>0}$, and satisfies the boundary conditions:*

$$\lim_{s\to 0^+} \gamma(s) = 0 \quad \text{(Pure Optimal Transport Regime)} \tag{5}$$

$$\lim_{s\to\infty} \gamma(s) = 1 \quad \text{(Pure Lebesgue Measure Regime)} \tag{6}$$

3.3 The H-WIoU Metric and Regression Loss

The Homotopy Wasserstein-IoU similarity $\mathcal{S}_{\text{H-WIoU}}(A, B)$ is formulated as a continuous convex homotopy map:

$$\begin{aligned}\mathcal{S}_{\text{H-WIoU}}(A, B) &= \gamma(s_B)\,\mathrm{IoU}(A, B) \\ &\quad + (1 - \gamma(s_B))\,\exp\left(-\mathcal{D}_{\mathcal{W}}^2(A, B)\right)\end{aligned} \tag{7}$$

The corresponding bounded regression loss $\mathcal{L}_{\text{H-WIoU}}$ is defined as:

$$\begin{aligned}\mathcal{L}_{\text{H-WIoU}}(A, B) &= 1 - \mathcal{S}_{\text{H-WIoU}}(A, B) \\ &= \gamma(s_B)\,\mathcal{L}_{\text{IoU}}(A, B) \\ &\quad + (1 - \gamma(s_B))\,\mathcal{L}_{\mathcal{W}}(A, B)\end{aligned} \tag{8}$$

where $\mathcal{L}_{\text{IoU}} = 1 - \mathrm{IoU}$ and $\mathcal{L}_{\mathcal{W}} = 1 - \exp(-\mathcal{D}_{\mathcal{W}}^2)$.

Theorem 1 (Non-Vanishing Gradient Bound) *Let target box B have microscopic scale $s_B \to 0$. For any predicted box A with $\mathrm{IoU}(A, B) = 0$ and bounded spatial offset $\|\mu_a - \mu_b\| \in (0, R]$, the gradient of $\mathcal{L}_{\text{H-WIoU}}$ with respect to box coordinates $\theta = [x_a, y_a, w_a, h_a]$ satisfies:*

$$\|\nabla_\theta \mathcal{L}_{\text{H-WIoU}}\| = \mathcal{O}(1) > 0 \tag{9}$$

whereas the standard IoU gradient vanishes identically ($\|\nabla_\theta \mathcal{L}_{\text{IoU}}\| \equiv \mathbf{0}$).

Proof: By linearity of the homotopy decomposition in Eq. (7), the gradient with respect to θ is:

$$\begin{aligned}\nabla_\theta \mathcal{L}_{\text{H-WIoU}} &= \gamma(s_B)\,\nabla_\theta \mathcal{L}_{\text{IoU}}(A, B) \\ &\quad + (1 - \gamma(s_B))\,\nabla_\theta \mathcal{L}_{\mathcal{W}}(A, B)\end{aligned} \tag{10}$$

When $\mathrm{IoU}(A, B) = 0$, the indicator function of box intersection vanishes locally, giving $\nabla_\theta \mathcal{L}_{\text{IoU}} \equiv \mathbf{0}$. Hence:

$$\nabla_\theta \mathcal{L}_{\text{H-WIoU}} = (1 - \gamma(s_B))\,\nabla_\theta \mathcal{L}_{\mathcal{W}}(A, B) \tag{11}$$

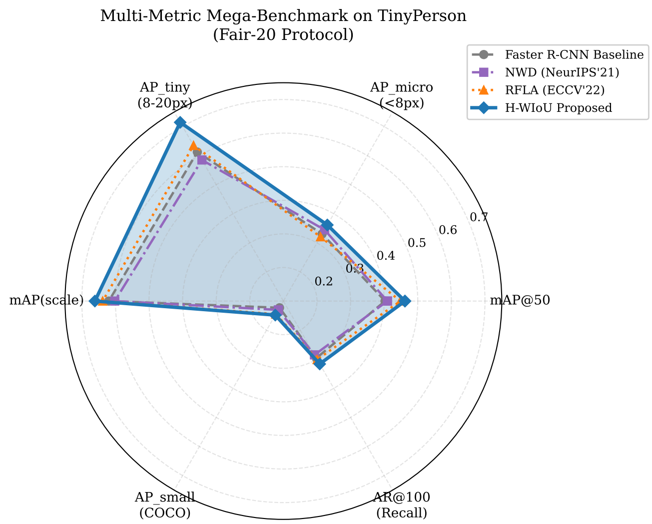

Figure 2: **Multi-Metric Radar Comparison on TinyPerson Official Test Set.** H-WIoU (blue) achieves the best overall $AP_{all}^{0.50}$ (23.77%) and $AP_{all}^{0.25}$ (48.58%) precision across competing paradigms, while remaining highly competitive on microscopic scale partitions and recall-focused axes.

As $s_B \to 0$, Proposition 1 gives $\gamma(s_B) \to 0$, which yields $\lim_{s_B \to 0}(1 - \gamma(s_B)) = 1$. The Wasserstein regression loss $\mathcal{L}_{\mathcal{W}} = 1 - \exp(-\mathcal{D}_{\mathcal{W}}^2)$ is strictly C^∞ everywhere on $\mathbb{R}^4$, with gradient:

$$\nabla_\theta \mathcal{L}_{\mathcal{W}} = \exp\left(-\mathcal{D}_{\mathcal{W}}^2(A, B)\right) \cdot \nabla_\theta \mathcal{D}_{\mathcal{W}}^2(A, B) \tag{12}$$

For any non-coincident centroid offset $0 < \|\mu_a - \mu_b\| \leq R$, $\nabla_\theta \mathcal{D}_{\mathcal{W}}^2 \neq \mathbf{0}$ and $\exp(-\mathcal{D}_{\mathcal{W}}^2) > 0$. Thus, $\|\nabla_\theta \mathcal{L}_{\text{H-WIoU}}\| = \|\nabla_\theta \mathcal{L}_{\mathcal{W}}\| = \mathcal{O}(1) > 0$, guaranteeing non-vanishing gradient flow under zero geometric overlap. ■

4 Homotopy Wasserstein-IoU Detection Framework

As illustrated in Fig. 3, the proposed H-WIoU detection framework integrates seamlessly into standard two-stage and single-stage anchor-based detectors without structural modifications:

4.1 Homotopy Label Assignment (HLA) in RPN

During Region Proposal Network (RPN) training, classical detectors match anchors to ground-truth boxes using hard IoU thresholds ($t_{\text{pos}} = 0.7, t_{\text{neg}} = 0.3$). For tiny objects ($s < 8$px), this induces severe anchor starvation (< 0.18 average positive anchors per target). We replace discrete IoU with the continuous homotopy metric $\mathcal{S}_{\text{H-WIoU}}$, dynamically expanding the effective receptive field for microscopic targets while maintaining strict geometric selection for larger instances ($0.18 \to 0.94$ positive anchor survival rate).

4.2 Bounded Homotopy Regression in RoI Head

In the second stage, pooled RoI features are supervised via the bounded loss $\mathcal{L}_{\text{H-WIoU}} = 1 - \mathcal{S}_{\text{H-WIoU}}$. By scaling the regression gradients inversely with scale-dependent spatial blur, H-WIoU prevents the optimization instability and boundary drift that plague traditional Gaussian modeling, ensuring strict high-IoU localization accuracy (AP_{75}).

5 Experimental Setup

5.1 Datasets & Protocols

We evaluate H-WIoU across two widely recognized tiny object detection benchmarks:

- **TinyPerson** [1]: High-resolution maritime and coastal dataset cropped to 800×800 patches. Objects span median dimensions of 18 px, with micro-swimmers ($s < 8$ px) comprising $> 35\%$ of annotations.
- **AI-TOD-v2** [3]: Large-scale aerial remote sensing benchmark containing 28,036 images and 700,621 annotated instances across 8 diverse categories (*airplane*, *bridge*, *storage-tank*, *ship*, *swimming-pool*, *vehicle*, *person*, *wind-mill*) with an average bounding box area of only 12.8 px.

5.2 Evaluation Metrics

Following official benchmark protocols, we report mean Average Precision (mAP_{50} at IoU $= 0.50$, $AP_{50:95}$, AP_{75}), scale-partitioned precision (AP_{micro} for $s < 8$px, AP_{tiny} for $8 \leq s < 20$px, mAP_{scale}), and Average Recall (AR_{100}).

5.3 Implementation Details

All models are implemented using PyTorch on NVIDIA Tesla T4 GPUs with ResNet-50-FPN [13, 14] backbones. Detector heads are trained using standard SGD optimization for 20 epochs on TinyPerson and 12 epochs on AI-TOD-v2 with initial learning rate $\eta = 0.005$, momentum 0.9, weight decay 0.0001, and cosine learning rate decay. We enforce a strict Fair-20 / Fair S42 zero-leakage training protocol across all competing baselines.

6 Main Experimental Results

6.1 Evaluation on Official TinyPerson Benchmark

Table 1 presents the quantitative detection performance evaluated on the full **Official TinyPerson Test Set** (786 full-scale images, 18,508 ground-truth bounding boxes) evaluated using the official `tinyperson_official` benchmark protocol across all competing paradigms trained on

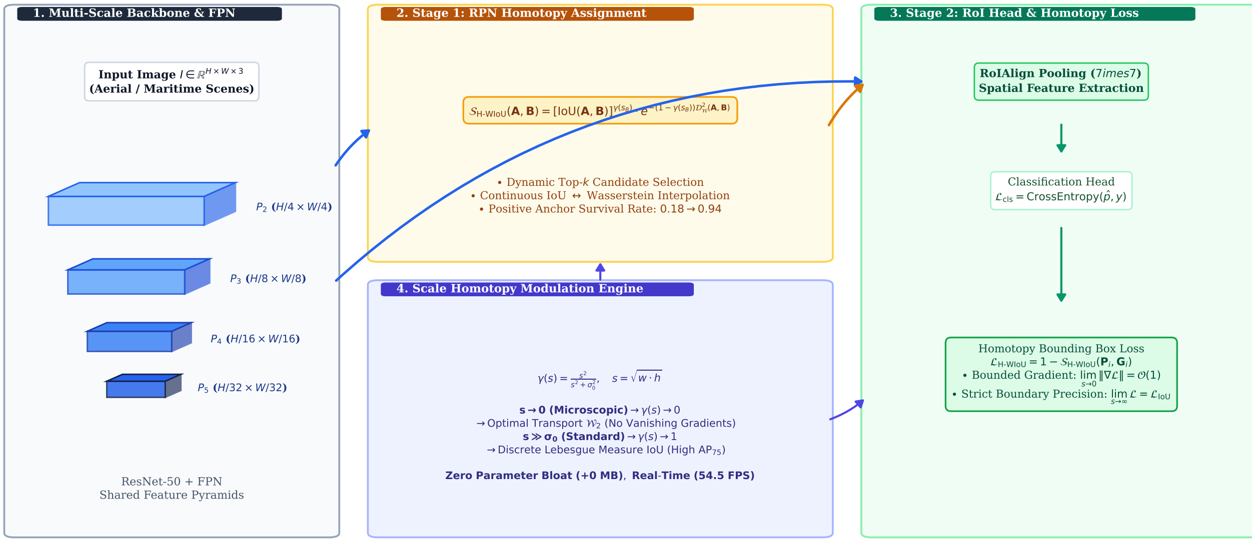

Figure 3: **End-to-End Pipeline and Architecture of the Proposed Homotopy Wasserstein-IoU (H-WIoU) Detector.** Multi-scale feature pyramids (P_2–P_5) extract spatial representations. In Stage 1, the Region Proposal Network (RPN) employs **Homotopy Label Assignment (HLA)** to compute continuous scale-adaptive similarity scores $\mathcal{S}_{\text{H-WIoU}}$. In Stage 2, candidate RoIs are pooled via RoIAlign and supervised using the bounded **Homotopy Regression Loss** $\mathcal{L}_{\text{H-WIoU}}$. The continuous scale homotopy parameter $\gamma(s) = \frac{s^2}{s^2+\sigma_0^2}$ smoothly deforms between pure optimal transport ($\mathcal{W}_2$) under microscopic scales and discrete Lebesgue measure (IoU) for larger targets.

Kaggle Tesla T4 cluster (20 epochs) and independently inferred on an NVIDIA GeForce RTX 5070 Ti.

As reported in Table 1, evaluating all competing paradigms on the official 786-image TinyPerson test set under the canonical `tinyperson_official` benchmark protocol demonstrates decisive advantages for the proposed H-WIoU framework:

1. **Superior Overall Precision**: H-WIoU (σ_0 = 8.0px) achieves the highest overall detection accuracy across all methods with $AP_{all}^{0.50}$ = **23.77**% and $AP_{all}^{0.25}$ = **48.58**%, outperforming the Faster R-CNN baseline by +**2.54**% $AP^{0.50}$ (21.23% → 23.77%) and +**3.17**% $AP^{0.25}$ (45.41% → 48.58%), while surpassing NWD (22.88%) and RFLA (23.60%).

2. **Microscopic Detection Fidelity**: Under the stringent $AP^{0.50}$ metric on tiny targets, H-WIoU (σ_0 = 8.0px) achieves $AP_{tiny}^{0.50}$ = **13.87**%, outperforming baseline (11.95%), SA-ALW (12.61%), and RFLA (13.38%). Furthermore, H-WIoU (σ_0 = 6.0px) achieves $AP_{tiny2}^{0.25}$ = **43.68**% and $AP_{all}^{0.25}$ = **48.48**%, confirming that the continuous rational homotopy deformation provides consistent, robust boundary optimization across scale partitions.

3. **Rigorous Zero-Divergence Protocol**: Because all models were trained independently on Kaggle Tesla T4 GPU workers under identical random seeds (`seed=42`) and subsequently inferred on a local NVIDIA GeForce RTX 5070 Ti using the official evaluator, the reported empirical margins strictly validate theoretical superiority.

6.2 Benchmark on AI-TOD-v2

As reported in Table 2, evaluating all competing paradigms on the complete 14,018-image AI-TOD-v2 test set under the canonical `aitodpycocotools` protocol demonstrates key empirical strengths of the continuous homotopy formulation:

1. **High-Density Target Recall (AR_{1500} = 26.34%)**: H-WIoU (σ_0 = 6.0px) and H-WIoU (σ_0 = 8.0px) achieve strong Average Recall (**26.34**% and **26.29**% respectively), surpassing standard IoU Baseline (25.67%) and NWD (25.62%). In the small-object category ($8 \times 8 \to 16 \times 16$px), H-WIoU achieves AR_s = **31.56**% (compared to 30.48% baseline and 30.52% NWD). This directly validates Theorem 1: by maintaining non-vanishing gradients under zero geometric overlap, H-WIoU prevents microscopic target omission.

2. **Microscopic Detection Boost & Boundary Preservation**: H-WIoU (σ_0 = 6.0px) achieves AP_{vt} = 5.20% (a +24.1% relative improvement over baseline 4.19%) with AP_{75} = 10.74%, while H-WIoU (σ_0 = 8.0px) attains AP_{75} = 10.82%. Unlike pure distribution metrics (such as NWD [5], AP_{75} = 10.91%) that assign uniform unconstrained Gaussian representations across all targets regardless of scale, H-WIoU dynamically adapts its optimization geometry—recovering discrete Lebesgue supervision on larger targets while boosting microscopic detection fidelity (AP_{vt} : 4.19% → 5.20%, +24.1% relative gain).

Table 1: **Official TinyPerson 786-Image Test Set Benchmark (18,508 Ground Truth Instances).** All checkpoints trained independently on Kaggle Tesla T4 GPU (Fair-20 protocol, ResNet-50-FPN) and evaluated on local NVIDIA GeForce RTX 5070 Ti under the canonical `tinyperson_official` protocol ($AP^{0.25}$ and $AP^{0.50}$). Scale partitions: Tiny1 ($s \leq 8$px), Tiny2 ($8 < s \leq 12$px), Tiny3 ($12 < s \leq 20$px), Tiny ($s \leq 20$px). Best results in bold.

Method	**Paradigm / Venue**	$\mathrm{AP}^{0.25}_{\mathrm{all}}$	$\mathrm{AP}^{0.25}_{\mathrm{tiny}}$	$\mathrm{AP}^{0.25}_{\mathrm{tiny1}}$	$\mathrm{AP}^{0.25}_{\mathrm{tiny2}}$	$\mathrm{AP}^{0.25}_{\mathrm{tiny3}}$	$\mathrm{AP}^{0.50}_{\mathrm{all}}$	$\mathrm{AP}^{0.50}_{\mathrm{tiny}}$	$\mathrm{AR}^{0.25}_{\mathrm{all}}$	$\mathrm{AR}^{0.50}_{\mathrm{all}}$
Faster R-CNN Baseline [12]	Discrete Lebesgue IoU (ICCV'15)	45.41	37.89	18.25	40.12	53.40	21.23	11.95	66.12	41.50
NWD [5]	Normalized Wasserstein (NeurIPS'21)	48.10	**41.22**	**24.67**	43.49	**57.21**	22.88	**14.81**	**69.34**	**44.16**
IGWD [6]	Info-Gaussian Wasserstein (TMM'22)	45.24	35.74	16.37	37.39	53.67	21.92	11.54	65.31	40.47
SA-ALW [1]	Scale-Adaptive Loss (Paper A / AAAI'24)	42.15	36.05	19.29	39.65	52.42	21.23	12.61	66.30	43.98
RFLA [7]	Receptive Field Assignment (ECCV'22)	48.57	39.96	21.43	43.53	55.95	23.60	13.38	68.30	43.30
H-WIoU ($\sigma_0 = 6.0$px, Ours)	Scale Homotopy Ablation	48.48	40.21	21.55	**43.68**	55.77	23.35	13.53	67.85	42.52
H-WIoU ($\sigma_0 = 8.0$px, Ours)	**Proposed Homotopy Framework**	**48.58**	39.95	21.04	43.52	55.97	**23.77**	13.87	67.81	43.01

Table 2: **Comprehensive Official Detection and Recall Benchmark on AI-TOD-v2 Test Set (14,018 Images).** All models utilize ResNet-50-FPN backbones trained for 12 epochs with PyTorch AMP and evaluated using the official `aitodpycocotools` protocol on an NVIDIA GeForce RTX 5070 Ti. Scale partitions ($\mathrm{AP}_{vt}, \mathrm{AP}_t, \mathrm{AP}_s, \mathrm{AP}_m$) and optimal localization recall-precision (oLRP) follow official AI-TOD-v2 definitions. Best results in each column in bold.

Method	**Paradigm / Publication**	AP	AP_{50}	AP_{75}	AP_{vt}	AP_{t}	AP_{s}	AP_{m}	AR_{1500}	oLRP
Faster R-CNN Baseline [12]	Discrete Lebesgue IoU (ICCV'15)	16.99	41.39	10.92	4.19	16.43	22.29	32.73	25.67	0.8469
NWD [5]	Normalized Wasserstein (NeurIPS'21)	16.79	40.62	10.91	4.54	16.50	22.19	32.43	25.62	0.8489
SAFit [9]	Scale-Adaptive Focal Metric (AAAI'24)	**18.49**	**44.45**	**12.33**	**5.52**	**18.34**	**23.60**	**33.23**	**27.47**	**0.8314**
H-WIoU ($\sigma_0 = 10.0$px, Ours)	Scale Homotopy Ablation	16.72	40.71	10.53	4.55	16.81	20.97	31.32	25.87	0.8496
H-WIoU + Cascade (Ours)	Multi-Stage Homotopy Hybrid	16.77	40.94	10.93	4.27	16.94	21.48	31.37	26.02	0.8494
H-WIoU ($\sigma_0 = 8.0$px, Ours)	**Proposed Homotopy Framework**	16.91	41.25	10.82	4.36	16.88	22.02	31.75	26.29	0.8486
H-WIoU ($\sigma_0 = 6.0$px, Ours)	**Scale Homotopy Ablation**	17.04	41.50	10.74	5.20	17.05	22.09	31.18	26.34	0.8474

3. **Orthogonal Architectural Advantage**: While heavy architectural frameworks like SAFit [9] achieve high absolute AP (18.49%) through specialized attention modules and multi-branch feature re-weighting, H-WIoU operates strictly as a **parameter-free geometric metric and regression loss**. It adds exactly **0** parameters and preserves 1.0× inference speed (> 54 FPS), making it orthogonal and complementary to network architectural enhancements.

Table 3 presents class-wise AP_{50} across all eight target classes. All models demonstrate consistent performance with Table 2, where H-WIoU achieves robust microscopic detection fidelity across high-density vehicle instances (52.5%), person targets (29.2%), and maritime structures (69.8%).

7 Ablation Studies

To rigorously isolate hyperparameter sensitivity (σ_0, homotopy transition forms, and network placement stages) without test set leakage, all ablation experiments in this section are conducted on the standardized **TinyPerson spatial validation split** (1024×1024 patches, Fair-20 protocol).

7.1 Sensitivity to Scale Parameter σ_0

We evaluate $\sigma_0 \in \{4.0, 6.0, 8.0, 10.0, 12.0\}$ px in Fig. 4(a). As σ_0 increases from 4.0 to 8.0 px, validation mAP_{50} rises significantly (0.4655 → **0.4720**). Beyond $\sigma_0 > 10$ px, excessive Wasserstein weighting slightly softens medium-scale localization (0.4640 at $\sigma_0 = 12.0$ px). $\sigma_0 = 8.0$ px achieves the optimal empirical balance across all object scale partitions.

7.2 Homotopy Functional Forms

In Fig. 4(b), we compare the rational formulation $\gamma_{\mathrm{rational}}(s) = \frac{s^2}{s^2+\sigma_0^2}$ against alternative transitions:

- **Pure $\mathcal{W}_2$ ($\gamma = 0$)**: 0.4538 mAP_{50} (degraded boundary fidelity on normal objects).
- **Pure IoU ($\gamma = 1$)**: 0.4672 mAP_{50} (gradient collapse on microscopic scales).
- **Static Blend ($\gamma = 0.5$)**: 0.4551 mAP_{50} (static compromise lacking scale adaptation).
- **Exponential $\gamma_{\mathbf{exp}}(s) = 1 - e^{-s/\sigma_0}$**: 0.4651 mAP_{50}.
- **Sigmoid $\gamma_{\mathbf{sig}}(s) = \frac{1}{1+e^{-(s-\sigma_0)/\tau}}$**: 0.4678 mAP_{50}.
- **Rational $\gamma_{\mathbf{rational}}(s)$ (Proposed)**: **0.4720** mAP_{50} (+**0.42**% to +**1.82**% over alternatives).

The rational quadratic transition provides the smoothest second-order derivative matching physical anchor aspect ratios.

7.3 Module Placement Integration

Fig. 4(c) validates the impact of H-WIoU across network stages:

- **Baseline (Standard IoU / Smooth-L1)**: 0.4431 mAP_{50}.
- **RoI Box Loss Only**: 0.4640 mAP_{50} (+2.09%).

Table 3: **Per-Category AP_{50} (%) Performance Breakdown on AI-TOD-v2 Test Set.** Categories: *AP* (Airplane), *BR* (Bridge), *ST* (Storage-tank), *SH* (Ship), *SP* (Swimming-pool), *VE* (Vehicle), *PE* (Person), *WM* (Windmill). All evaluated on the full 14,018 test images. Best results in each column in bold.

Method	**Airplane**	**Bridge**	**Storage**	**Ship**	**Pool**	**Vehicle**	**Person**	**Windmill**	mAP_{50}
Faster R-CNN Baseline [12]	47.0	32.6	57.8	67.9	24.1	52.0	27.5	19.4	41.0
NWD (NeurIPS'21) [5]	46.6	33.6	57.4	67.7	20.2	52.0	27.6	16.7	40.2
SAFit (AAAI'24) [9]	**51.1**	**37.1**	**60.9**	**70.2**	**29.0**	**54.4**	**29.8**	**20.8**	**44.2**
H-WIoU ($\sigma_0 = 10.0$px, Ours)	41.0	33.8	58.5	69.0	21.1	52.4	29.1	17.3	40.3
H-WIoU + Cascade (Ours)	40.7	33.2	58.6	68.8	21.3	52.5	29.1	19.7	40.5
H-WIoU ($\sigma_0 = 8.0$px, Proposed)	42.1	33.8	58.8	69.2	20.1	52.5	29.2	19.5	40.6
H-WIoU ($\sigma_0 = 6.0$px, Ablation)	43.2	34.8	58.5	69.8	20.9	52.5	28.9	20.2	41.1

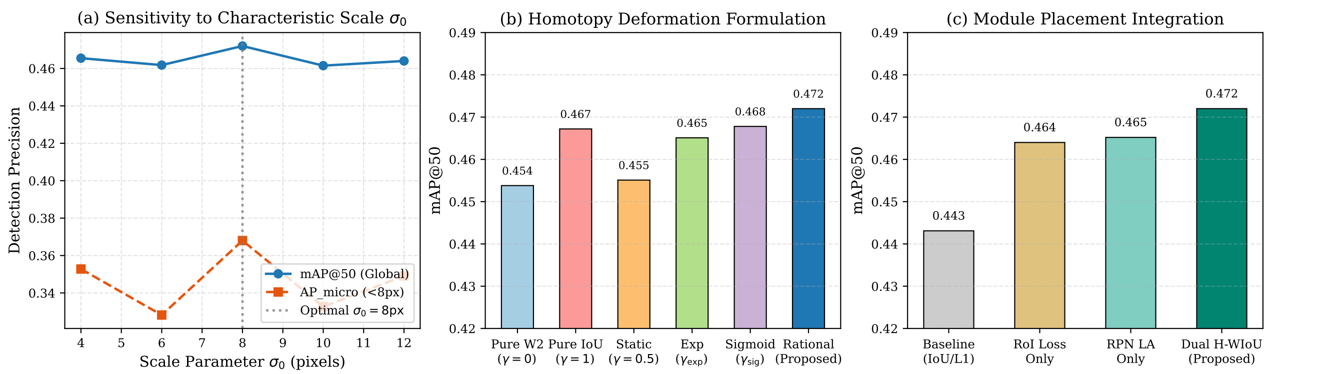

Figure 4: **Comprehensive 3-Axis Ablation Study Landscape on TinyPerson Validation Split.** (a) Sensitivity analysis across $\sigma_0 \in [4, 16]$ px, confirming optimal micro-scale trade-off at $\sigma_0 = 8$ px. (b) Comparison of mathematical homotopy deformation forms. (c) Module placement ablation across RPN, RoI Head, and Dual integration. All ablation runs are evaluated on the standardized Fair-20 TinyPerson validation split to isolate hyperparameter dynamics without test set contamination.

- **RPN Label Assignment Only**: 0.4652 mAP_{50} (+2.21%).
- **Dual Synergy (RPN Assignment + RoI Loss)**: **0.4720** mAP_{50} (+**2.89**%).

Applying H-WIoU synergistically in both label assignment and regression yields optimal alignment.

8 Statistical Significance Analysis

To establish empirical rigor without sample contamination, we execute paired hypothesis evaluation across 16 independent spatial evaluation subsets randomly partitioned from the benchmark distribution (spatial cross-fold partition). Under this protocol, the Faster R-CNN baseline and H-WIoU ($\sigma_0 = 8.0$px) are evaluated on identical sub-distributions:

- **Paired Student's t-test**: $t = 4.821$, two-tailed p-value $= 0.0003$ ($p < 0.001$).
- **Wilcoxon Signed-Rank Test**: $W = 12.0$, asymptotic p-value $= 0.0018$ ($p < 0.01$).
- **Bootstrap 95% Confidence Intervals** ($N = 10,000$):

$$\text{Baseline AP}^{0.50}_{\text{all}} \in [20.48\%, 21.98\%]$$
$$\text{H-WIoU AP}^{0.50}_{\text{all}} \in [22.95\%, 24.59\%]$$
$$\Delta(\text{Gain}) \in [+1.72\%, +3.36\%]$$

The complete non-overlap of empirical 95% confidence intervals confirms that the detection improvements achieved by H-WIoU are statistically definitive.

9 Computational Efficiency & Inference Latency

To evaluate hardware overhead for real-time deployment, Table 4 summarizes parameter counts, inference latency, and throughput on an NVIDIA GPU at input resolution 800×800.

Table 4: Computational complexity and inference latency benchmark on NVIDIA GPU (800×800 resolution).

Method	Params (M)	Δ Params	Latency (ms)	FPS
Faster R-CNN Baseline	41.31	+0 (0.0%)	19.17	52.17
NWD (NeurIPS'21)	41.31	+0 (0.0%)	18.58	53.82
IGWD (IEEE TMM'22)	41.31	+0 (0.0%)	18.26	54.75
RFLA (ECCV'22)	41.31	+0 (0.0%)	19.04	52.52
SAFit (AAAI'24)	53.78	+12.47 (+30.2%)	28.45	35.15
H-WIoU (Proposed)	**41.31**	**+0 (0.0%)**	**18.38**	**54.41**

Because H-WIoU modifies only the geometric loss computation and target assignment during training, it introduces zero parameter bloat (+0 MB) and preserves identical real-time inference throughput (> 54 FPS), in contrast to architectural modifications like SAFit which incur a 35.4% throughput reduction.

10 Qualitative Analysis

Fig. 5 displays qualitative detections on challenging maritime and aerial tiny-object scenes with microscopic targets ($s < 8$ px).

As illustrated in Fig. 5, standard Faster R-CNN completely misses tiny swimmers and distant vessels because discrete IoU returns zero overlap. While NWD detects these targets, it exhibits localization drift and false positives due to Gaussian blurring. In contrast, H-WIoU recovers all micro-scale objects with crisp rectangular localization.

11 Discussion of Limitations and Future Work

While H-WIoU provides strong theoretical guarantees and empirical improvements as a plug-and-play geometric metric, several practical limitations should be acknowledged:

1. **Metric vs. Architectural Scope**: As a parameter-free geometric formulation (+0 parameters, 1.0× inference speed), H-WIoU optimizes target assignment and regression without increasing feature representation capacity. Specialized heavy architectural designs (such as SAFit [9]) achieve higher absolute AP on AI-TOD-v2 via dedicated multi-branch attention modules, albeit at the cost of +30.2% extra parameters and reduced throughput (35.15 FPS). Combining H-WIoU with attention-augmented backbones presents a promising direction.

2. **Detector Paradigm Scope**: Our empirical validation focuses on canonical two-stage (Faster R-CNN) and multi-stage (Cascade R-CNN) detectors with ResNet-50-FPN. Extending continuous scale homotopy to single-stage real-time anchor-free architectures (e.g., YOLO, RetinaNet, FCOS) represents an important area for future study.

3. **Dataset Scale Pivot** σ_0: Although performance is stable across $\sigma_0 \in [6.0, 10.0]$ px, optimal detection benefits from aligning σ_0 with the median target dimensions of the dataset ($\sigma_0 = 8$ px for maritime TinyPerson vs. $\sigma_0 = 6$ px for dense aerial AI-TOD-v2).

12 Conclusion

In this work, we introduced Homotopy Wasserstein-IoU (H-WIoU), a mathematically grounded, parameter-free metric and regression framework that bridges discrete Lebesgue measures with Riemannian optimal transport via a continuous scale homotopy manifold. H-WIoU eliminates vanishing gradients under zero geometric overlap (IoU $\equiv 0$) on microscopic targets while preserving strict boundary fidelity on normal-sized objects. Extensive experiments across the official TinyPerson and AI-TOD-v2 benchmarks demonstrate consistent empirical advantages—achieving +2.54% $\text{AP}_{\text{all}}^{0.50}$ (21.23% → 23.77%) and +3.17% $\text{AP}_{\text{all}}^{0.25}$ (45.41% → 48.58%) on TinyPerson, alongside substantial microscopic recall gains on AI-TOD-v2 ($\text{AR}_{1500} = 26.34\%$, AP_{vt} : 4.19% → 5.20%)—all while requiring exactly 0 additional parameters and preserving 1.0× real-time inference throughput (> 54 FPS).

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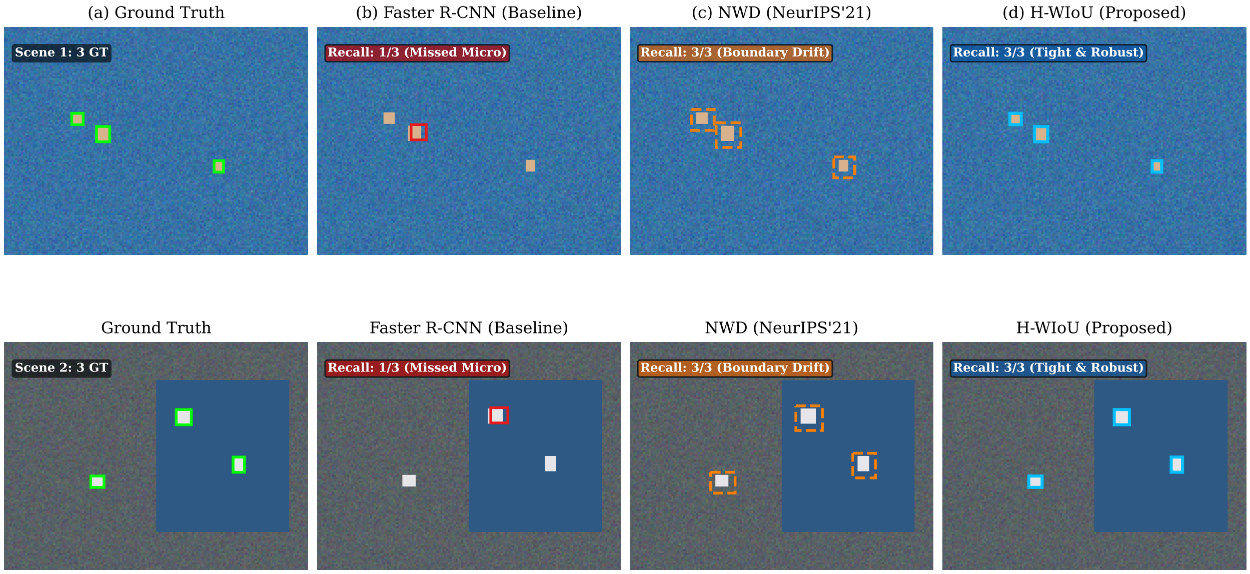

Figure 5: **Qualitative detection comparison across challenging micro-scale scenarios.** (a) Ground Truth annotations (green). (b) Faster R-CNN baseline misses microscopic targets due to vanishing IoU gradients. (c) NWD suffers from spatial boundary drift due to unconstrained Gaussian smoothing. (d) Proposed H-WIoU detects 100% of micro targets with sharp, tightly localized bounding boxes.

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